

KENNETH W. BINGHAM

AI Systems Architect • Senior Software Engineer • Full-Stack Platform Owner

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GitHub: github.com/kenbin64

Portfolio: ButterflyFX.us • KensGames.com • TTLRecall.com • DimensionSOS.com

PROFESSIONAL SUMMARY

AI Systems Architect and Senior Software Engineer with more than ten years of enterprise software development experience and a highly accelerated, immersive specialization in advanced AI systems engineering. Designs deterministic agent frameworks, directive logic systems, geometric reasoning structures, and multi model orchestration pipelines. Builds disciplined, non drifting AI using logical directives, geometry as data, fractal imaginary number boundaries, truth table routing, and schema validated pipelines. Owns platforms end to end from model directives to Ubuntu VPS deployment. Known for transforming AI from reactive prediction into structured, focused, and reliable reasoning systems.

AI Systems Engineering: deterministic agent loops, directive engineering, model directives, multi model routing, RAG, vector search, evaluation harnesses, hallucination reduction, drift control, long context reliability

Advanced Reasoning Architecture: geometry as data, fractal imaginary number boundaries, minimal surface routing, manifold based reasoning, directive discipline frameworks

Models: Llama 3.1, Qwen 2.5, DeepSeek V3, DeepSeek Coder, Mistral, Mixtral, Phi 3, StarCoder2, Llava, GPT 4.1, Claude 3.5, Gemini 1.5, Copilot

Backend Engineering: Java 8 to 17, .NET 6, Python, Spring Boot, REST APIs, microservices, deterministic state engines

Infrastructure: Ubuntu VPS admin, Docker, Kubernetes, NGINX, PM2, SSL, GitHub Actions, Azure Pipelines, CI and CD automation

Data Systems: SQL Server, Oracle, MySQL, Postgres, SQLite, MarkLogic, vector stores, schema design, indexing, optimization

Frontend and 3D: Angular, React, Three.js, WebGL, WebGPU, interactive 3D visualization

ADVANCED AI SYSTEMS ENGINEERING

Private AI Ecosystem (Self Hosted on Ubuntu VPS)

Built a sovereign AI platform on a self managed Ubuntu VPS with NGINX, PM2, SSL, domain routing, firewall hardening, and secure SSH workflows.

Hosted local LLMs including Llama, Qwen, DeepSeek, Mistral, Phi 3, StarCoder2, and Llava using Ollama, LM Studio, and custom Python runners.

Integrated frontier cloud models including GPT 4.1, Claude 3.5, Gemini 1.5, and Copilot through a directive controlled orchestration layer.

Designed a multi model router using truth table constraints, minimal surface boundaries, and deterministic fallback logic.

Applied geometric reasoning boundaries using fractal imaginary numbers to stabilize model behavior and reduce hallucinations.

VS Code Multi Agent Development Environment

Engineered a VS Code AI control surface using Continue.dev, Tabby, and custom extensions.

Built specialized agents including Architect Agent, Coder Agent, Verifier Agent, Router Agent, Memory Agent, Scenario Agent, and Dimensional Agent.

Enabled parallel reasoning, cross model verification, and directive driven orchestration.

Created directive logic that enforces discipline, focus, and bounded reasoning across all agents.

Directive and Manifold Based AI Control

Authored model directives defining allowed behavior, schema, retry budgets, and reasoning boundaries.

Built the Schwarz D routing system using minimal surface geometry for tier selection.

Created the Tiresias manifold for state lineage, ancestry preservation, and deterministic reasoning flow.

Implemented truth table and decision tree routing to ensure frontier models are used only when required.

Applied geometry as data and fractal imaginary number points to create stable, non drifting reasoning paths.

PROFESSIONAL EXPERIENCE

Software Developer — DTSI

Apr 2024 to Present • Hill Air Force Base

Modernized mission critical Air Force systems using Java, .NET, and SQL Server.
Built deterministic test harnesses reducing regression risk across legacy workflows.

Operated in DoD cleared, high security environments with strict change control.

Software Developer — Humana

Feb 2022 to Feb 2024 • Remote

Developed .NET 6 services, HL7 and FHIR pipelines, and SQL optimizations.

Built Python ETL automation and Linux admin workflows reducing manual cycles.

Software Developer — Systems Implementers

Apr 2021 to Dec 2021 • Hill Air Force Base

Built Java and Spring Boot services for aircraft maintenance and asset tracking.

Software Developer — Gwinnett County School District

Feb 2020 to Aug 2020 • Georgia

Built automated test frameworks in Java and Selenium improving QA coverage.

Software Developer — Norfolk Southern Railroad

Aug 2019 to Jan 2020 • Atlanta, GA

Improved locomotive optimization software and handled real time data processing.

Software Developer — Finicity (via STG)

Nov 2017 to Aug 2018 • Murray, UT

Built Java based financial data extraction and secure API integrations.

Java Programmer — Wex

Feb 2017 to Sep 2017 • Ogden, UT

Delivered enterprise web portals and document management workflows.

Java Programmer — CUProdigy

Apr 2015 to Feb 2017 • Utah

Built secure banking applications in Java, Spring, Hibernate, and MySQL.

INDEPENDENT PLATFORMS (SOLE ARCHITECT AND DEVELOPER)

ButterflyFX.us — AI reasoning research platform including Ultimate AI Directive, Schwarz D routing, and Tiresias manifold.

KensGames.com — deterministic 3D game engine with manifold compiler and unified rule substrate.

TTLRecall.com — verified scenario generation platform with two production systems built in thirty minutes.

DimensionSOS.com — stateless deterministic AI architecture hub.

EDUCATION

B.S. Computer Science and Software Engineering

SELECTED WRITING AND OPEN SOURCE

Substack (@kbingh) — AI architecture and dimensional computing

GitHub — manifold compiler, routing systems, scenario tooling

The Ultimate AI Directive

Schwarz D AI Routing System